

INDUWARA JAYAWICKRAMA

BSc. (Hons) in Computer Engineering (UG)

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[Induwara Jayawickrama](#)



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PROFILE

I am an enthusiastic, self-motivated and reliable computer engineering undergraduate with a strong passion for software engineering. I thrive in both collaborative team environments and when working independently. I am highly adaptable to new and challenging situations and I approach problems with a proactive mindset and a keen attention to detail. I perform well under pressure, consistently meeting tight deadlines while maintaining quality and precision. My goal is to contribute meaningfully to innovative projects while continuing to learn and grow in the software engineering domain.

EDUCATIONAL QUALIFICATIONS

Bachelor of Science in Computer Engineering

2022 to Present

University of Jaffna, Sri Lanka

Relevant Coursework: Software Engineering, Object-Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Advanced Networking, Embedded Systems Design, Computer Architecture, Robotics and Automation, Control Systems, Modeling in Operational Research.

WORK EXPERIENCE

Software Engineer Intern | Sri Lanka Telecom– Colombo, Sri Lanka

11/2025 to 05/2026

- Served as primary Developer and Team Leader, managing daily team tasks, aligning development goals, and leading the end-to-end delivery of the project.
- Developed full stack enterprise solutions using the PERN stack (PostgreSQL, Express.js, React.js, Node.js) and Flutter, maintaining high performance and code reliability.
- Monitored DevOps workflows and facilitated deployments by providing infrastructure teams with critical system requirements, configuration details, and deployment specifications.
- Acted as the main technical point of contact for clients, translating business requirements into actionable development tasks and managing stakeholder expectations.
- Collaborated closely with Business Analysts (BAs) to gather initial requirements and assist in authoring comprehensive technical documentation and system specifications.
- Partnered with QA engineers to streamline bug tracking, accelerate issue resolution, and maintain rigorous application quality standards.

PROJECTS

Bookify – Service Booking Web Application 

01/2025 – 04/2025

Spring Boot | React.js | MySQL | RESTful APIs

Bookify is a full stack service booking web application developed as a group project. It connects consumers with service providers, allowing users to browse services, view provider profiles, check availability and make bookings. Service providers can register on the platform, manage their profiles and receive bookings.

System Automation Project – Facility Management Department

01/2026-05/2026

React.js | Node.js | Express.js | PostgreSQL | Tailwind CSS | JWT | RESTful APIs

Engineered a full-stack automation platform to manage regional evaluations and systemic task tracking. Developed a highly responsive, modern administration dashboard using React, styled with Tailwind CSS, and integrated Recharts for dynamic analytical data visualization. Designed a secure backend architecture with Express.js, utilizing PostgreSQL for centralized database management. Implemented rigorous JSON Web Token (JWT) authentication for role-based access routing and automated system alert notifications using Nodemailer.

Hiring Vehicle Management System (HVMS)

11/2025-05/2026

Flutter | Node.js | Express.js | MySQL | Firebase | BLoC | RESTful APIs

Developed a cross-platform vehicle logistics and management application utilizing Flutter for the frontend with the BLoC pattern for efficient state management. Built a scalable backend utilizing Node.js and Express.js, MySQL for flexible data handling. Integrated Firebase Authentication for secure access control and implemented interactive geolocation features

using flutter_map. Utilized Syncfusion DataGrid and built-in CSV/Excel parsing functionalities to generate comprehensive transport summary reports.

Face2Face – Real-Time Video Chat Application

02/2025 – 04/2025

WebRTC | Socket.IO | JavaScript | HTML | CSS | React.js | Node.js | Express.js | Railway | Firebase | Vercel | GitHub

Developed a browser-based peer-to-peer video chat application using WebRTC and Socket.IO. The platform enables users to join rooms, share video/audio streams, exchange text messages and toggle screen sharing and recording functionality. The project emphasized real-time communication, media stream handling and frontend backend integration without relying on third-party video services

Accident Detection and Prevention System

02/2025 – 04/2025

ESP32 | Ultrasonic Sensor | GSM | GPS | Flame Sensor | C++ | Node.js | Google Maps API | GitHub

Developed a smart embedded system that detects potential road accidents and initiates emergency communication. Utilized ultrasonic sensors and ESP32 for real-time obstacle and pedestrian detection. Integrated GSM and GPS modules to send automated alerts with location data to emergency services. Focused on pre-accident prevention and post-accident response, ensuring driver awareness and timely intervention through a web app interface.

Blockchain-Enhanced Security for Lightweight IoT Embedded Systems (Research)

01/2025 – Present

IoT | Blockchain | ECDSA | AES-128 | SHA-256 | MQTT | Python | Mininet | SQLite | Wireshark

Researching a lightweight blockchain framework to secure IoT systems in resource-constrained environments. Designed a three-layer architecture using AES-128 for encryption, ECDSA for authentication and a simplified Proof of Stake consensus. Implemented secure communication via MQTT and developed validator logic in Python. Evaluated system performance through real hardware testing and simulation using Mininet and Wireshark, focusing on metrics such as energy consumption, latency, and attack resilience across heterogeneous IoT scenarios.

TECHNICAL SKILLS

Programming Languages | Java, C++, C, Python, PHP, HTML, CSS, JavaScript, MATLAB
Frameworks | React.js, Node.js, Express.js, Spring Boot, Flutter
Databases | MySQL, MongoDB, PostgreSQL
Tools & Platforms | Google Colab, Jupyter Notebook, Git, Postman, Vercel, Firebase, Railway
Embedded Systems | Arduino, Embedded C
Network and Security | Wireshark, WebRTC, Cisco Packet Tracer

SOFT SKILLS

- Problem Solving
 - Communication Skills
- Critical Thinking
 - Teamwork
- Flexibility
 - Creativity
- Leadership
 - Adaptability

CERTIFICATES

- Lead Yourself Certificate of Completion Personality Development Program
- Time Management (From Chegg)
- YarInsight 2.0 - Summer School on Full Stack Web Development with the MERN Stack

VOLUNTEERING

Web Master of IEEE PES Student Chapter - University of Jaffna

04/2024 – Present

Managed and maintained the Chapter's website, organized virtual events via Zoom and Google Meet and created Google Forms for registrations, feedback, etc. Ensured seamless online communication and timely updates of Chapter activities. Supported the team with technical aspects of digital outreach.

President of the Media Committee | Engineering Student Union - University of Jaffna

08/2025 – 08/2026

Led and managed the overall operations of the Media Committee, coordinating digital outreach strategies and supervising the production of promotional multimedia content for Union events. Spearheaded a team of creative designers and sub-editors to maintain consistent content quality, enhance the Union's digital presence across-platforms, and ensure clear, timely updates on all official student activities.

Sub Editor of Engineering Student Union- University of Jaffna

08/2023 – 08/2024

Created and curated engaging posts for various events, ensuring clear communication and promotion of Union activities across digital platforms. Collaborated with the team to maintain consistent content quality and timeliness.

COMPETITIONS

IEEEExtreme 18.0	108th all island and 6th place of SBC UOJ	2024
CodeRally 5.0	Rank 12	2024
SLRC /Line Following Robotics Competition		2024
MoraXtreme		2023, 2024

PROFESSIONAL MEMBERSHIPS

Student Member: Institution of Engineers Sri Lanka (IESL), Membership ID: S-29538	2023 - Present
Student Member: IEEE Sri Lanka Section	2023 – Present

REFERENCES

Dr. (Eng.) A. Kaneswaran Dean, Senior Lecturer Grade II, Faculty of Engineering, University of Jaffna kanesh@eng.jfn.ac.lk +94 77 204 2863	Mr. M.Giridaran Manager – Software Development, Sri Lanka Telecom PLC, Colombo 01 mgiri@slt.com.lk +94 71 536 4876
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